

CPM Live Submission Packet

Bonded-template DEM reveals strength- and topology-dependent fracture-event sequences in packed brittle ceramic pebbles

Core Package

Field	Value
Target journal	Computational Particle Mechanics
Submission route	ScienceDirect / Editorial Manager live submission route
Main upload package	submission_packages/ computational_particle_mechanics_upload_ready.zip
Optional blinded package	submission_packages/ computational_particle_mechanics_blinded_review_optional.zip
Reduced reproducibility package	submission_packages/repair_submission_package.zip
Missing author e-mails	7
Public candidate e-mail lookup	docs/cpm_author_email_public_lookup_20260704.md
Public candidate e-mails	4

Upload Sequence

Step	Item	File or value	Required	Note
1	Manuscript	01_manuscript.pdf	yes	Use the full manuscript unless the live system asks for a blinded file.
2	Highlights	02_highlights.docx	yes	Five highlights are also available in the editorial fields document.
3	Graphical abstract	03_graphical_abstract.png or 03_graphical_abstract.tif	if requested	PDF and SVG backups are included for editable artwork.
4	Declaration of competing interest	04_declaration_of_competing_interest.docx	yes	No competing interests declared.
5	Cover letter	05_cover_letter.docx	yes	Targeted to Computational Particle Mechanics.
6	Author details and contributions	06_author_emails_and_contributions.docx	yes	Use with completed coauthor e-mail sheet.
7	LaTeX source	07_latex_source.zip	if requested	Editable source for the full manuscript.
8	Editorial paste fields	08_editorial_submission_fields.docx	support	Use for title, abstract, keywords, data and code availability.
9	Main figures	09_main_figures.zip	if requested	Contains separate PDF, PNG and SVG figure files.
10	Author e-mail completion sheet	10_author_email_completion_sheet.docx / .csv	external metadata	Seven coauthor e-mails remain missing in local records.
11	Optional blinded manuscript	computational_particle_mechanics_blinded_review_optional.zip	only if requested	Use for double-anonymized review workflow if the system asks for a blinded file.
12	Reduced	repaired_submission_p	support	Use for data/code

	reproducibility package	ackage.zip		repository or supplemental support if requested.
--	-------------------------	------------	--	--

Paste Fields

Journal

Computational Particle Mechanics

Source: START_HERE_CPM_SUBMISSION.md

Submission route

ScienceDirect / Editorial Manager live submission route

Source: docs/cpm_official_submission_guide_alignment_20260704.md

Article type

Research article

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Title

Bonded-template DEM reveals strength- and topology-dependent fracture-event sequences in packed brittle ceramic pebbles

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Keywords

brittle ceramic pebble; bonded-particle DEM; packed bed; particle breakage; force network; fracture-event sequence; bond strength

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Abstract

Packed brittle pebbles can lose mechanical integrity through a small number of localized particle fractures before bed-wide fragmentation is visible. Here, we develop a bonded-template discrete-element workflow that inserts 500-subparticle brittle templates into random packed beds with no pre-loading bond loss and records parent-particle fracture events during compression. A nominal 1 mm Li₄SiO₄ template is calibrated against single-pebble crush-load and split-fragment constraints. Accepted 100-particle bed windows retain 493,500 initial internal bonds, measurable load transfer and a connected native force graph before fracture loading. The completed dataset contains four localized cracking sequences, three zero-loss controls and an eleven-row material-response table, including a six-case strength matrix across two cracking geometries. The early-cracking geometry keeps first localized bond loss at 19 micrometres as strength is reduced from nominal to 0.25x, with endpoint bond loss increasing from 10 to 15 only at 0.25x. The synchronous-cracking geometry shifts first localized bond loss from 39 micrometres at nominal strength to 19 micrometres at 0.50x and 0.25x, again increasing endpoint bond loss to 15 only at 0.25x. These results show that fracture onset in finite packed-particle windows is controlled by the combination of local force-path topology and bonded-particle strength. The workflow converts packed-bed compression from bulk force reporting into a source-data-backed event sequence with particle identity, bond-loss increment and native force-network state.

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Highlights

- Bonded templates resolve parent-particle fracture events inside random packed beds. (83 characters)
- Intact 500-subparticle templates are inserted without pre-loading bond loss. (76 characters)
- Event tables record displacement, bond-loss increment and damaged-particle identity. (84 characters)

- Force-network state variables separate zero-loss and localized-cracking endpoints. (82 characters)
- A bond-strength matrix shows geometry-dependent shifts in fracture onset. (73 characters)

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Data availability

The processed event tables, native force-network summaries, material-response tables, figure source data, manuscript figures and post-processing scripts supporting this study are provided in a reduced reproducibility package with a checksum manifest. The package has been deposited in a persistent repository and is available at <https://doi.org/10.5281/zenodo.20687351>. Very large raw restart files and full local-bond dump histories are retained as larger archives and can be provided on reasonable request subject to repository file-size limits and institutional storage constraints.

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Code availability

The template-generation, run-control, figure-generation and post-processing scripts are included in the reduced reproducibility package under `scripts/`, with representative DEM input files under `simulations/`, and are mirrored in the GitHub repository [wangjianfttt/Mother_particle_bonded_dem_research](https://github.com/wangjianfttt/Mother_particle_bonded_dem_research). The citable archived version uses the same repository DOI, <https://doi.org/10.5281/zenodo.20687351>. The package reproduces manuscript-level processed data tables, figures and source-data matrices from the included CSV tables and representative inputs. The local DEM executable, very large restart files and full raw local-bond dump histories are retained outside the compact package.

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Declaration

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Funding / acknowledgements

This work was supported by the Anhui Provincial Natural Science Foundation (2408085QA030), the Science Foundation of the Institute of Plasma Physics, Chinese Academy of Sciences (DSJJ-2025-08), the Anhui Intelligent Mine Technology and Equipment Engineering Research Center (AIMTEERC202307), the China Post-doctoral Science Foundation under Grant Number 2024M753266, the Excellent Research and Innovation Team of Anhui Province, China (2022AH010052), the Scientific Research Foundation for High-level Talents of Anhui University of Science and Technology, China (2021yjrc51), and the Collaborative Innovation Program of Hefei Science Center, CAS, China (2019HSC-CIP006).

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Corresponding author

Jian Wang, wjfttt@mail.ustc.edu.cn

Source: manuscript/computational_particle_mechanics_editorial_fields.md

Missing Author E-mails

- Siyu Wang
- Hang Zhang
- Ming-Zhun Lei
- Wei Wen
- Qi-Gang Wu
- Gang Shen
- Haishun Deng

Public Candidate E-mail Lookup

These entries are public candidates for corresponding-author confirmation; they do not change the formal missing-email count.

Author	Candidate e-mail	Status	Source
Ming-Zhun Lei	leimz@ipp.ac.cn	Public candidate; confirm before final submission	https://pmc.ncbi.nlm.nih.gov/articles/PMC8234535/
Wei Wen	wenwei@ipp.ac.cn	Public candidate; confirm before final submission	https://www.ipp.cas.cn/yjs/xsc/202207/P02025011577706650358.pdf
Gang Shen	shenganghit@163.com	Public candidate; confirm before final submission	https://journals.sagepub.com/doi/10.1177/09544062241299521
Haishun Deng	269469122@qq.com	Public candidate; confirm before final submission	https://pmc.ncbi.nlm.nih.gov/articles/PMC11637116/

Final External Actions

- Fill seven coauthor e-mail addresses if required by the live submission system.
- Confirm public candidate e-mails for Ming-Zhun Lei, Wei Wen, Gang Shen and Haishun Deng before using them in the live system.
- Confirm article type/category in the live system.
- Preview the system-generated PDF before final submit.